

predictor of Claude usage (relevance) but being measured independently, is expected to be uncorrelated with sampling noise in our AUI estimates (validity). As noted above, states with more workers in high-Claude-usage roles do tend to have systematically higher usage per capita.

The 2SLS estimates imply modestly slower convergence:  $\hat{\beta} \approx 0.89$  unweighted and  $\hat{\beta} \approx 0.86$  when weighting by each state's working-age population. However, these estimates are less precise, and only the former is statistically distinguishable from 1 at the 10% level. Despite implying a slower convergence than OLS, the 2SLS estimates still imply rapid diffusion: just four to five years for the log deviation of each state's AUI to shrink by 90%.

That said, our estimates are based on just three months of data. And while the 2SLS specification may help address sampling noise, considerable uncertainty remains. We will revisit this question of the pace of diffusion in future reports.

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<sup>1</sup> As with previous reports, all our analysis is based on privacy-preserving analysis. Throughout the report we analyze a random sample of 1M conversations from Claude.ai Free, Pro and Max conversations (we also refer to this as “consumer data” since it mostly represents consumer use) and 1M transcripts from our first-party (1P) API traffic (we also refer to this as “enterprise data” since it mostly represents enterprise use). Both samples come from November 13, 2025 to November 20, 2025. We continue to manage data according to our privacy and retention policies, and our analysis is consistent with our terms, policies, and contractual agreements. For 1P API data, each record is a prompt-response pair from our sample period which in some instances is mid-session for multi-turn interactions.

<sup>2</sup> The share of conversations on Claude.ai that were classified into neither automation nor augmentation categories fell from 3.9% to 3.0%.

<sup>3</sup> See, for example, [Kalanyi et al \(2025\)](#): “Second, as the technologies mature and the number of related jobs grows, hiring spreads geographically. This process is very slow, taking around 50 years to disperse fully.”

patterns do not hold. Task success and human education show opposite relationships globally versus within the US.

The near-perfect correlation between human and AI education years underscores that how users prompt Claude shapes how it responds. Combined with the finding that higher-usage countries engage Claude more collaboratively, this suggests that the skills required to use AI well may themselves be unevenly distributed.

By measuring the characteristics of conversations with Claude, we find important relationships with broader economic factors such as human capital. These relationships may help predict labor market outcomes and inform a smooth transition to an AI-enabled economy that will require different skillsets.

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<sup>1</sup> For privacy reasons, our automated analysis system filters out any cells—e.g., countries, and (country, task) intersections—with fewer than 15 conversations and 5 unique user accounts. For bottom-up request clusters, we have an even higher privacy filter of at least 500 conversations and 250 unique accounts.

<sup>2</sup> Data in this section covers 1 million Claude.ai Free, Pro and Max conversations from November 13 to 20, 2025, randomly sampled from all conversations in that period. We then excluded content that was flagged as potential trust and safety violations. The unit of observation is a conversation with Claude on Claude.ai, not a user, so it is possible that multiple conversations from the same user are included, though our [past work](#) suggests that sampling conversations at random versus stratified by user does not yield substantively different results. Aggregate geographic statistics at the country and US state level were assessed and tabulated from the IP address of each conversation. For geolocation, we use ISO-3166 codes since our provider for IP geolocation uses this standard. International locations use ISO-3166-1 country codes, US state level data use ISO-3166-2 region codes, which include all 50 US states and Washington DC. We exclude conversations originating from VPN, anycast, or hosting services, as determined by our IP geolocation provider.

crosses the horizontal 50% success line at 3.5 hours, suggesting that API calls attain a 50% success rate for tasks that are 3.5 hours. The analogous time estimate in METR's software engineering benchmark is 2 hours for Sonnet 4.5 and about 5 hours for Opus 4.5. (The data in this report predates the release of Opus 4.5.)

Claude.ai data tells a different story. Success rates decline far slower as a function of task length. Extrapolating using the linear fit, Claude.ai would hit a 50% success rate at about 19 hours. This may reflect how multi-turn conversation effectively breaks complex tasks into smaller steps, with each turn providing a feedback loop that allows users to correct course.

It's worth noting that a fundamental difference from the METR setting is selection. METR constructs a benchmark where a fixed set of tasks is assigned to models. In our data, users choose which tasks to bring to Claude. This means observed success rates reflect not just model capability but also user judgment about what will work, the cost of setting up the problem for Claude, and the expected time savings if the task succeeds.

If users avoid tasks they expect to fail, for example, observed success rates will overstate true capability on the full distribution of potential tasks. This selection likely operates on both platforms, but in different ways: API customers select for tasks amenable to automation, while Claude.ai users select for tasks that could benefit from iteration. Also due to this selection effect, there's no guarantee that more performant models would show improvement in this plot, because users may respond to new models by providing more challenging presentations of otherwise similar O\*NET tasks.

Controlled benchmarks like METR's measure the frontier of autonomous capability. Our real-world data can measure the *effective* task horizon, reflecting a mix of model capabilities and user behavior, and expanding beyond coding tasks. Both approaches find that AI can be effective for tasks requiring hours of human work.